2026 S.T. Yau High School Science Award (Asia)
Research Outline

	Team Member	Team Member	Team Member
Name	Palaash Gang		
School	Indus International School, Pune		
City, Country	Pune, India		
Registration No.	Chem-151		

	Supervising Teacher
Name	Yogita Hastak Menon
Position	Head of Senior School
School/Institution	Indus International School, Pune
City, Country	Pune, India

Overcoming the Irving-Williams Preference in a Galloyl-Ossein Biosorbent through Experimentally and Computationally Driven Hemidirected Lead(II) Selectivity

Abstract

Highly selective recovery of trace Pb(II) from mixed-metal wastewaters remains a critical environmental challenge. However, conventional non-specific biosorbents typically lose finite oxygen-donor sites to transition metals like Cu(II) and Zn(II), dictated by the Irving-Williams stability sequence. Because Pb(II) lies outside these classical stability sequences selectively capturing it requires tuning donor geometry, polarizability, and desolvation thermodynamics rather than relying on simple charge density. We hypothesize that a galloyl-rich oxygen-donor environment can exploit these distinct properties to actively redirect selectivity toward Pb(II).

To test this, ossein (demineralized fish-scale collagen) was functionalized via aqueous grafting with tannic acid (~5.5 wt%) to immobilize vicinal-trihydroxy (galloyl) pockets. ATR-FTIR confirmed functionalization (aromatic C=C, galloyl-associated ester bands) while preserving the collagen Amide I-III architecture. Single-metal Langmuir isotherms (pH 5.0, below Pb(II) hydroxide precipitation) yielded maximum capacities of Pb(II) > Cu(II) > Zn(II) (40.11, 25.38, and 16.86 mg/g, respectively). While the transition metals retained their typical Irving-Williams ordering, Pb(II) notably outranked them. In equal-mass ternary competition (50 mg/L each)—where Pb(II) faced a >3-fold molar disadvantage—the sorbent captured 71.6% of the Pb(II), compared to just 41.0% Cu(II) and 26.4% Zn(II). The resulting selectivity factors ($\alpha(Pb/Cu) = 3.63$, $\alpha(Pb/Zn) = 7.03$) significantly exceeded the unfunctionalized ossein control ($\alpha(Pb/Cu) = 1.24$, $\alpha(Pb/Zn) = 2.33$). Furthermore, continuous-flow fixed-bed columns processed ~45 bed volumes before 10% Pb(II) breakthrough, and counter-current acid regeneration concentrated the eluted lead 11–14x, retaining 80.1% capacity across three

cycles.

Density-functional theory (DFT) modeled the localized galloyl coordination environment. Calculated solution-phase binding free energies ($\Delta G°\_bind$ = -145.2, -110.4, and -85.6 kJ/mol for Pb(II), Cu(II), and Zn(II), respectively) accurately reproduced the experimentally observed Pb > Cu > Zn preference. Energy-decomposition analysis (EDA) revealed the inadequacy of a purely electrostatic rationale: electrostatic stabilization (ΔE_elstat) was highest for the more compact Cu(II) center, which would incorrectly predict an inverted selectivity. However, while Pb(II) exhibited lower absolute electrostatic stabilization, its orbital interaction fraction ($f\_orb = \Delta E\_orb / (\Delta E\_elstat + \Delta E\_orb) = 0.38$) was the highest among the metals, indicating a significantly more covalent interaction.

Natural Orbitals for Chemical Valence (NOCV) analysis localized this covalency to electron donation from the galloate π and oxygen-lone-pair manifold into the polarizable 6p orbitals of Pb(II)—a charge-transfer pathway less accessible to the harder, more contracted Cu(II) and Zn(II) acceptors. Crucially, the galloate pocket uniquely accommodates the stereochemically active $6s^2$ lone pair of Pb(II), enforcing a hemidirected coordination geometry. Harder transition metals, by contrast, bind more ionically and hydrate strongly ($\Delta G°\_hyd \approx -2100$ kJ/mol for Cu(II) vs. -1481 kJ/mol for Pb(II)), thus incurring substantially larger desolvation penalties. This desolvation-controlled binding was corroborated by experimental thermodynamics, which confirmed an entropy-driven adsorption mechanism ($\Delta H°$ = +15.2 kJ/mol, $\Delta S°$ = +136.5 J/(mol·K)). Ultimately, this study recasts Pb(II) selectivity as a synergistic coordinative effect driven by orbital covalency, hemidirected geometry, and favorable desolvation—demonstrating how fundamental computational molecular design translates directly into a high-performance, recoverable environmental material.

Keywords: Lead(II) Selectivity, Density-Functional Theory (DFT), Coordination Chemistry, Biosorption, Multi-Metal Competition, Desolvation, Waste Valorization, Heavy Metal Remediation

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